Multi-Agent Systems

Multiple rounds of tool calling to interact with the world

- Multi-agent systems use multiple LLMs with developer-defined personas to give each agent a certain "expertise"

- These roles are defined by a human expert pre-hoc

- These manually defined multi-agent systems are powerful, but generalize across domains poorly

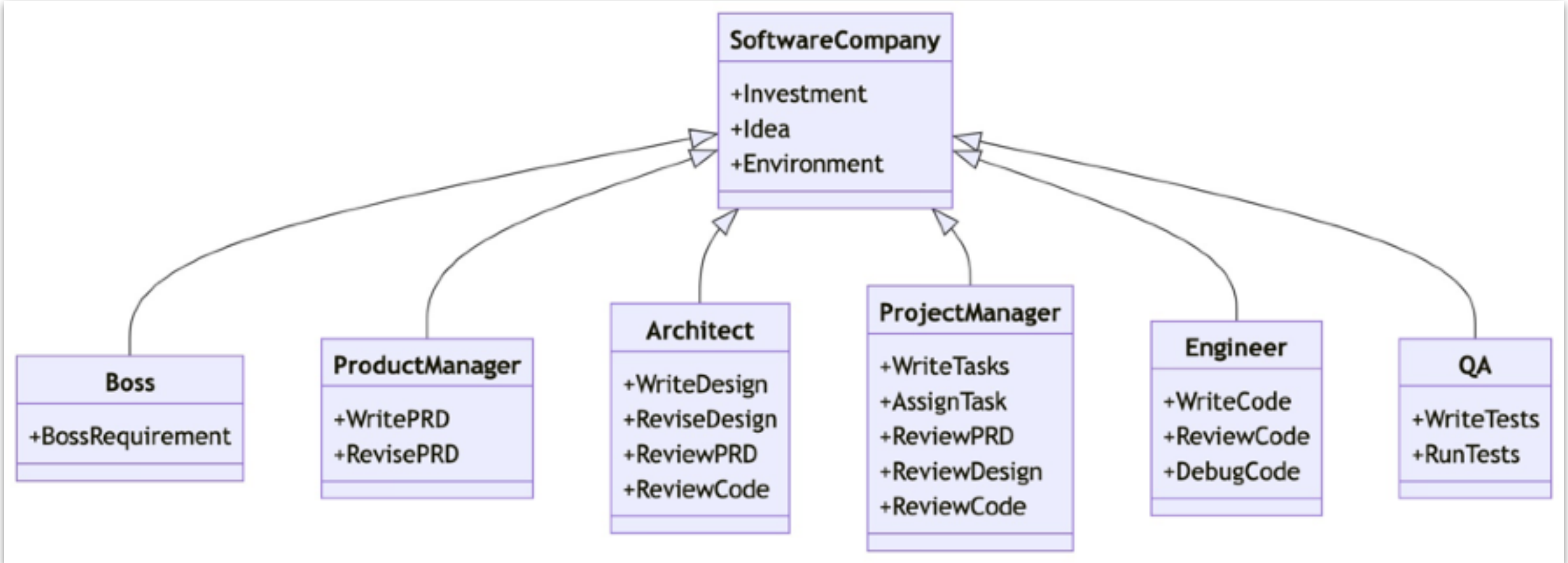
SoftwareCompany
+Investment
+Idea
+Environment
Boss
+BossRequirement
ProductManager
+WritePRD
+RevisePRD
Architect
+WriteDesign
+ReviseDesign
+ReviewPRD
+ReviewCode
ProjectManager
+WriteTasks
+AssignTask
+ReviewPRD
+ReviewDesign
+ReviewCode
Engineer
+WriteCode
+ReviewCode
+DebugCode
QA
+WriteTests
+RunTests

Figure from MetaGPT (Hong et al., 2024)

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- **How can we design multi-agent systems to decompose and delegate complex tasks without human-defined graphs?**

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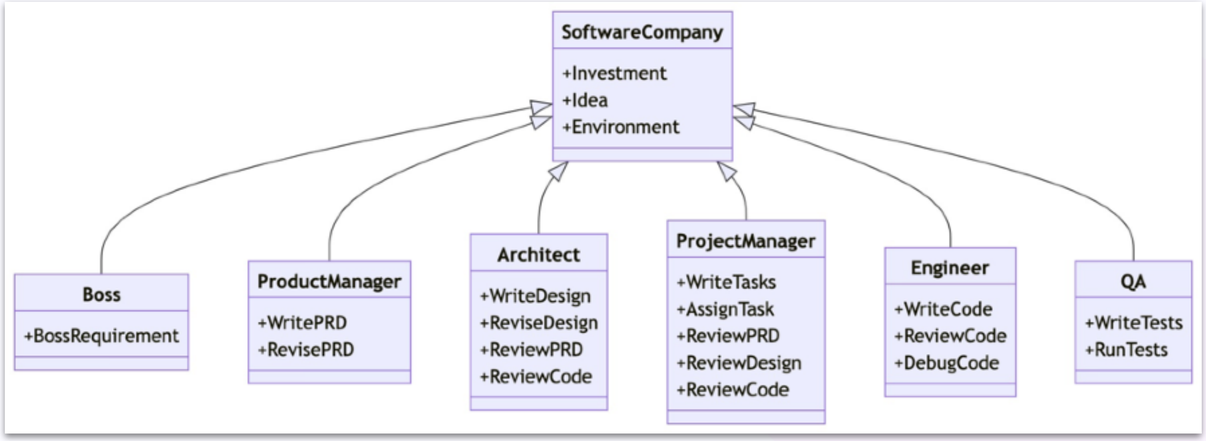

Figure from MetaGPT (Hong et al., 2024)

Recursive Delegation

A framework for recursive multi-agent systems